Reg. No. | | | | | | | | | | | [17AE53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2019

AIRCRAFT STRUCTURES - II

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.
ii. Missing data may be suitably assumed.

PART - A

1 a. Explain different states of equilibrium of a column and define the critical load. **03**

b. A 1.5m long column has a circular cross section with 30 mm diameter has both ends *clamped*. Discuss stability criteria of the column when it is subjected to the following axial loads: **04**

i. 130 kN

ii. 140 kN

iii. 150 kN (Assume E=200677MPa.)

c. Obtain critical buckling load expression for a *fixed- pinned* column and locate the inflexion point and mark the effective column length. **07**

2 a. Using approximate method based on total potential (Π_p) find critical load for a cantilever column using the shape function $v(z)=Az^2(3L-z)$ where A is an arbitrary constant, v is displacement and z is measured along the length L. **06**

b. Compare the results with the exact solution. **08**
For an initially bent column derive an expression for the deflection at the midpoint. Draw load vs deflection curve and hence define critical buckling load.

3 a. Define beam-column action with an example. Name any two aircraft components where beam column action is present. **04**

b. Figure 3, illustrates part of a typical wing spar of thin walled round alloy steel tube acting as a beam column. It is supported at pin A and by the inclined strut BC at point C. Determine the safety of the beam column when carrying a uniform lateral load of 2N/mm. Assume ultimate bending modulus of rupture of the given tube as 1200 MPa and E=210 GPa. **10**

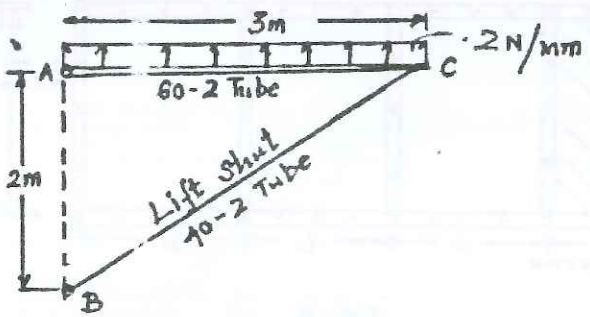

Fig. 3

4 Obtain an expression for critical buckling stress σ_{cr}, of a simply supported rectangular plate *a/b* under uniaxial compression load per unit length, N_x. **14**

Graphically show the variation of buckling coefficient kc *vs* a/b and state all important observations.

PART - B

5 **a.** For the following plate configuration find the critical buckling coefficient for a/b≥4 **03**

i. Loaded edges *simply supported* and unloaded edges *simply supported* and free.

ii. Loaded edges *clamped* and unloaded edges *clamped* and simply supported.

iii. Loaded edges *simply supported* and unloaded edges *clamped* and free.

b. Explain instability behavior of a typical composite shape open section column by illustrating the compressive stress distribution on the cross section. **04**

c. For the **Z** section shown in Fig. 5 estimate the crippling stress. **07**

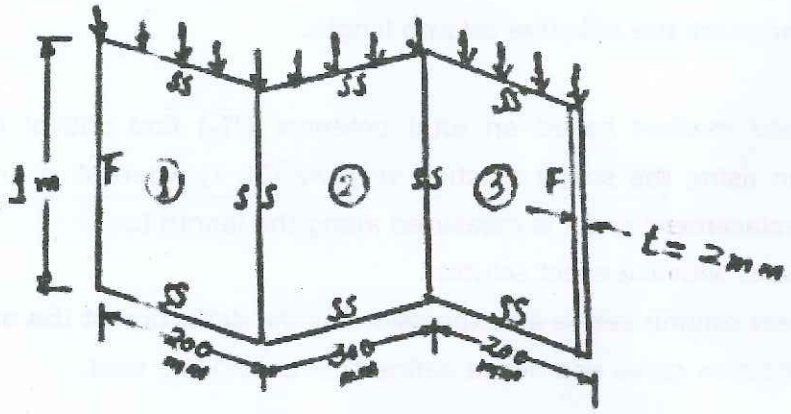

Fig. 5

6 **a.** Explain pure diagonal tension field in a flange -web beam with a neat sketch. **06**

b. A tension field beam is shown in Fig.6. The cross-section area and moment of inertia of the flanges are 500 mm^2 and 1000 mm^4. The thickness of the sheet is 2 mm. Determine the failure status of the wing beam and modify the design if there is a failure in the flanges or the stiffeners. **08**

Assume, σ_{cu} = 400 N/mm^2, E=70GPa & v =0.3. Diagonal Tension angle= 43°.

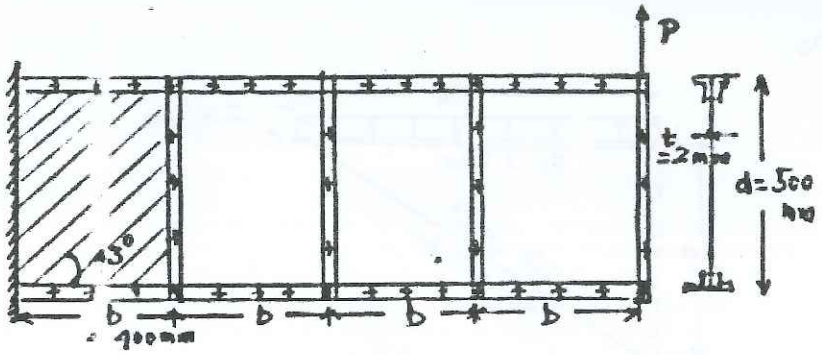

Fig. 6

7 **a.** Classify monocoque cylinders based on geometry parameter **Z** and explain their buckling behavior. **04**

b. A monocoque circular cylinder has the following geometry: **10**

Length L= 2400 mm; Diameter, d=2400 mm and wall thickness, t = 1 mm.

It is subjected to the following load system:

Axial compression= 40kN; Bending Moment=20kNm and Torsional moment=30 kNm.

Check for the buckling failure of the monocoque cylinder under the combined loading system. Material used: Al. Alloy Ec= 74 GPa; v =0.3.

8 **a.** An idealized Fuselage section having radius of 2.4m with longitudinal stringers is shown in Fig.8. The area of each stringer is 100mm^2. The skin thickness is 1.2mm. The fuselage frame spacing, a=4b. Where b is the curved width between the stringers. **14**

The fuselage section is subjected to the following load system:

Bending moment: My= 14000 k Nm; Torsion: Mx = 15000 k Nm;

Identify the critical skin panel in the fuselage section and check its ultimate strength.

All material is of aluminum alloy with E=72 GPa and v=0.3

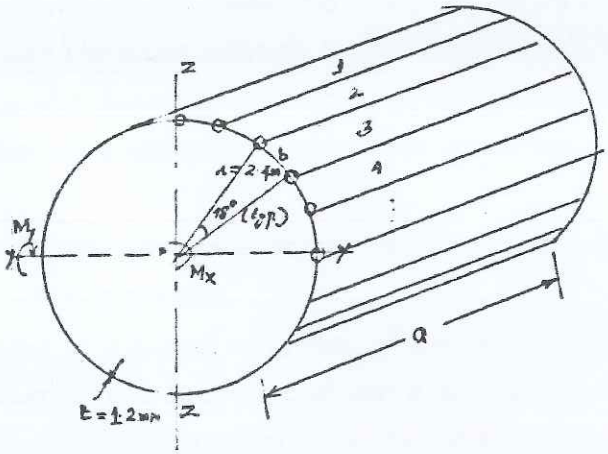

Fig. 8